

## N. Bora Keskin

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### Education

Stanford University

Ph.D., Graduate School of Business, Jun. 2012

- Academic Area: Operations, Information & Technology
- Dissertation Title: "Dynamic Pricing with Demand Model Uncertainty"
- Advisors: J. M. Harrison and A. Zeevi
- GPA: 6.00/6

Boğaziçi University

B.Sc., Mathematics, Jun. 2007

B.Sc., Industrial Engineering, Jun. 2006

- Overall GPA: 4.00/4

### Employment

Duke University, Fuqua School of Business

Associate Professor (tenured), 2023–present

Associate Professor (without tenure), 2020–2023

Assistant Professor, 2015–2020

University of Chicago, Booth School of Business

Assistant Professor, 2012–2015

McKinsey & Company

Business Analyst, 2006–2007

### Selected Honors and Awards

Winner, MSOM Young Scholar Prize, 2024

Winner, Lanchester Prize, 2019

Winner, Triangle Impact Challenge, 2021

Markov Lecture Discussant, INFORMS Applied Probability Society, 2023

Invited Speaker, MSOM Teaching Innovation Session, 2025

Winner, Best Paper Award, Digital Supply Chain and Supplier Diversity Conference, 2022

Winner, INFORMS Service Science Best Paper Competition, 2022

Winner, INFORMS Data Mining Best Paper Competition, 2020

Honorable Mention, INFORMS Junior Faculty Interest Group (JFIG) Paper Competition, 2018

Distinguished Service Award

*Management Science*, 2019

Meritorious Service Award

*Management Science*, 2016, 2017, 2018, 2020

*Manufacturing & Service Operations Management (M&SOM)*, 2019, 2020

*Operations Research*, 2012

STAR Visitor, Netherlands Organization for Scientific Research, 2015

### Publications

1. Bayesian dynamic pricing policies: Learning and earning under a binary prior distribution, with J. M. Harrison & A. Zeevi (2012), *Management Science*, 58(3), 570–586.
2. Dynamic pricing with an unknown demand model: Asymptotically optimal semi-myopic policies, with A. Zeevi (2014), *Operations Research*, 62(5), 1142–1167.

3. Chasing demand: Learning and earning in a changing environment, with A. Zeevi (2017), *Mathematics of Operations Research*, 42(2), 277–307.
  - Lead article.
  - Winner, **Lanchester Prize**, 2019.
4. On incomplete learning and certainty-equivalence control, with A. Zeevi (2018), *Operations Research*, 66(4), 1136–1167.
5. Dynamic selling mechanisms for product differentiation and learning, with J. R. Birge (2019), *Operations Research*, 67(4), 1069–1089.
6. Discontinuous demand functions: Estimation and pricing, with A. V. den Boer (2020), *Management Science*, 66(10), 4516–4534.
7. Personalized dynamic pricing with machine learning: High-dimensional features and heterogeneous elasticity, with G.-Y. Ban (2021), *Management Science*, 67(9), 5549–5568.
  - Honorable mention, INFORMS Junior Faculty Interest Group (JFIG) Paper Competition, 2018.
  - Finalist, INFORMS Data Mining Best Paper Competition, 2019.
8. Competition between two-sided platforms under demand and supply congestion effects, with F. Bernstein & G. DeCroix (2021), *M&SOM*, 23(5), 1043–1061.
9. Dynamic learning and market making in spread betting markets with informed bettors, with J. R. Birge, Y. Feng, & A. Schultz (2021), *Operations Research*, 69(6), 1746–1766.
  - Preliminary version in proceedings of the 2019 ACM Conference on Economics and Computation (EC '19).
  - Featured in *Chicago Booth Review*.
10. Data-driven dynamic pricing and ordering with perishable inventory in a changing environment, with Y. Li & J. Song (2022), *Management Science*, 68(3), 1938–1958.
11. Impact of information asymmetry and limited production capacity on business interruption insurance,” with Y.-M. Kao & K. Shang (2022), *Management Science*, 68(4), 2824–2841.
12. Dynamic pricing with demand learning and reference effects, with A. V. den Boer (2022), *Management Science*, 68(10), 7112–7130.
13. Selling quality-differentiated products in a Markovian market with unknown transition probabilities, with M. Li (2024), *Operations Research*, 72(3), 885–902.
14. To interfere or not to interfere: Information revelation and price-setting incentives in multiagent learning environments, with J. R. Birge, H. Chen, & A. Ward (2024), *Operations Research*, 72(6), 2391–2412.
  - Featured in *Chicago Booth Review*, *China Business Knowledge*, and *UNSW BusinessThink*.
15. The blockchain newsvendor: Value of freshness transparency and smart contracts, with C. Li & J. Song (2025), *Management Science*, 71(8), 6666–6682.
  - Winner, Best Paper Award, Digital Supply Chain and Supplier Diversity Conference, 2022.
  - Winner, Fan Favorite Award, YinzOR Conference, 2022.
  - Second place, POMS-HK Best Student Paper Competition, 2023.
  - Honorable mention, *OR/MS Tomorrow* Mini Poster Competition, 2022.
  - Honorable mention, INFORMS Minority Issues Forum Poster Competition, 2023.
  - Featured in *Duke Fuqua Insights*, *INFORMS News Releases*, and *The Duke Daily*.
16. Markdown policies for demand learning with forward-looking customers, with J. R. Birge & H. Chen (2025), *Operations Research*, 73(5), 2550–2566.
  - Featured in *Chicago Booth Review*.

17. Data-driven clustering and feature-based retail electricity pricing with smart meters, with Y. Li & N. Sunar (2025), *Operations Research*, 73(5), 2636–2660.
  - Winner, INFORMS Data Mining Best Paper Competition, 2020.
  - Winner, INFORMS Service Science Best Paper Competition, 2022.
  - Invited to the 2022 Conference on AI, ML, & Business Analytics.
  - Featured in *Rethinc. Labs News*.

### **Papers Under Review or Revision**

18. Data-driven population tracking in service systems, with F. Bernstein, A. Mersereau, M. Wood, & S. Ziya.
  - Winner, **Triangle Impact Challenge**, 2021 (originated from the winning proposal).
  - Featured in *Duke Fuqua Insights* and *UNC Kenan-Flagler News & Stories*.
19. Order stacking in on-demand delivery platforms, with J. Scott & R. Swinney.
20. The nonstationary newsvendor: Data-driven nonparametric learning, with X. Min & J. Song.
21. Bayesian dynamic pricing and subscription period selection with unknown customer utility, with Y.-M. Kao & K. Shang.
22. Dynamic learning for joint pricing, advertising, and inventory management, with H. Gurkan & R. Parker.
23. Show-up profiles for scheduled services: Estimation and applications, with F. Bernstein, A. Mersereau, M. Wood, & S. Ziya.
  - Winner, **Triangle Impact Challenge**, 2021 (originated from the winning proposal).
  - Featured in *UNC Kenan-Flagler News & Stories*.
24. Green ride-hailing: Impact on environment and consumers, with Y. Li & N. Sunar.
  - Featured in *Johns Hopkins Carey News*.
25. Feature-based scheduling and dynamic learning with a large backlog, with C. Zhang.
26. Innovator’s edge in supply chain transparency and food waste reduction, with C. Li & J. Song.
27. A model-free geotemporal clustering approach for pandemic projection and medical resource planning: Insights from COVID-19, with X. Min & J. Song.
28. Optimal dynamic pricing with demand model uncertainty: A squared-coefficient-of-variation rule for learning and earning.
29. Deep learning for visual advertising on digital platforms, with Y. Li & J. Song.
30. Feature selection for promotion: Learning to avoid information overload, with C. Li & M. Li.
  - Winner, Best Student Paper Award, INFORMS Workshop on Data Science, 2024.

### **Works in Progress**

31. Endogenous state transitions in transfer learning and dynamic pricing, with J. Song & S. Wang.
32. Learning and earning for congestion-prone service systems, with P. Afèche.

### **Book Chapters**

33. Dynamic pricing with demand learning: Emerging topics and state of the art, with A. V. den Boer (2022), *The Elements of Joint Learning and Optimization in Operations Management*, Springer Series in Supply Chain Management, vol 18. Springer, Cham.
34. Dynamic pricing and demand learning in nonstationary environments, with A. V. den Boer (2022), *The Elements of Joint Learning and Optimization in Operations Management*, Springer Series in Supply Chain Management, vol 18. Springer, Cham.

### **Invited Research Talks (in alphabetical order)**

Amazon, 2019  
Bilkent University, Department of Industrial Engineering, 2014  
Boğaziçi University, Department of Industrial Engineering, 2019  
Boğaziçi University, Department of Management, 2020  
Boston College, Carroll School of Management, 2023  
Columbia University, Graduate School of Business, 2012  
Cornell University, Johnson Graduate School of Management, 2019  
Cornell University, School of Operations Research and Information Engineering, 2012  
Cornell University, School of Operations Research and Information Engineering, 2014  
Deloitte, 2022  
Duke University, Fuqua School of Business, 2012  
Duke University, Fuqua School of Business, 2015  
Duke University, Fuqua School of Business, 2019  
Duke University, Fuqua School of Business, 2025  
Georgetown University, McDonough School of Business, 2021  
Johns Hopkins University, Carey Business School, 2019  
Koç University, Graduate School of Business, 2022  
London Business School, 2016  
London Business School, 2022  
McGill University, Bensadoun School of Retail Management, 2021  
McGill University, Desautels Faculty of Management, 2014  
METU, Department of Industrial Engineering, 2021  
MIT, Data Science Lab, 2020  
MIT, Data Science Lab, 2022  
MIT, Sloan School of Management, 2012  
MIT, Sloan School of Management, 2014  
North Carolina State University, Poole College of Management, 2024  
Northwestern University, Kellogg Graduate School of Management, 2011  
Northwestern University, Kellogg Graduate School of Management, 2012  
NYU, Stern School of Business, 2012  
NYU, Stern School of Business, 2023  
Purdue University, Krannert School of Management, 2022  
Queen's University, Smith School of Business, 2017  
Queen's University, Smith School of Business, 2022  
Stanford University, Graduate School of Business, 2011  
Stanford University, Graduate School of Business, 2022  
Stanford University, Graduate School of Business, 2025  
Syracuse University, Whitman School of Management, 2021  
University of British Columbia, Sauder School of Business, 2022  
University of California at Riverside, School of Business, 2023  
University of Chicago, Booth School of Business, 2012  
University of Chicago, Booth School of Business, 2015  
University of Chicago, Harris School of Public Policy, 2015  
University of Illinois at Chicago, College of Business Administration, 2018  
University of Illinois at Urbana-Champaign, Gies College of Business, 2014  
UNC-Chapel Hill, Kenan-Flagler Business School, 2013  
UNC-Chapel Hill, Department of Statistics and Operations Research, 2014  
University of Southern California, Marshall School of Business, 2016

University of Texas at Austin, McCombs School of Business, 2020  
University of Texas at Dallas, Naveen Jindal School of Management, 2021  
University of Toronto, Rotman School of Management, 2015  
University of Twente, Department of Stochastic Operations Research, 2015  
University of Utah, David Eccles School of Business, 2024  
VU Amsterdam, Stochastics Department, 2015  
Washington University in St. Louis, Olin Business School, 2019  
Yale University, School of Management, 2025

**Invited Research Conferences** (in alphabetical order)

Conference on AI, ML, & Business Analytics: 2022  
Digital Supply Chain & Supplier Diversity Conference: 2022  
EURO-INFORMS Joint International Meeting: 2013  
INFORMS Annual Meeting: 2009–2024  
INFORMS Applied Probability Society Conference: 2013, 2015, 2017  
INFORMS Data Mining & Decision Analytics Workshop: 2020, 2021, 2024  
INFORMS Data Science Workshop: 2020, 2021, 2024  
INFORMS Revenue Management & Pricing Conference: 2010–2022  
Marketplace Innovation Workshop: 2017, 2019, 2021  
MSOM Conference: 2013, 2015, 2017, 2021, 2024, 2025  
MSOM Interface of Finance, Operations, & Risk Management (iFORM) SIG Workshop: 2018, 2019  
MSOM Service Management SIG Workshop: 2011, 2024  
MSOM Supply Chain Management SIG Workshop: 2023  
MSOM Teaching Innovation Session: 2025  
POMS Annual Conference: 2016, 2019  
Stochastic Networks, Applied Probability, and Performance (SNAPP) Seminar: 2025  
Supply Chain Finance & Risk Management Workshop: 2022  
Workshop in Information & Learning in Decisions and Operations: 2021  
YA/EM Congress on Operations Research & Industrial Engineering: 2021

**Teaching Experience**

Duke University, Fuqua School of Business

Value Chain Innovation in Business Processes (Daytime MBA and Executive MBA), 2019–2025  
– Course Evaluations: 6.41–6.98/7 (across sections and years)  
Supply Chain Management (Daytime MBA and Executive MBA), 2015–2019  
– Course Evaluations: 6.03–6.75/7 (across sections and years)  
Dynamic Pricing & Revenue Management (Ph.D.), 2016, 2018, 2020, 2023, 2025  
Emerging Topics in Operations Management (Ph.D.), 2017

University of Chicago, Booth School of Business

Operations Management: Business Process Fundamentals (MBA), 2013–2015  
– Course Evaluations: 4.63–4.76/5 (across sections and years)  
Dynamic Pricing & Revenue Management (Ph.D.), 2014

Stanford University, Graduate School of Business

Fundamentals of Operations, Information, and Technology (Ph.D.), 2009–2010

**Professional Activities and Service**

Associate Editor

*Management Science*, 2019–present

*Operations Research*, 2021–present

*Management Science*, Special Issue on Data-Driven Prescriptive Analytics

## Senior Editor

*Production and Operations Management*, 2024–present

## INFORMS Revenue Management & Pricing Section

Award Committee Chair, Jeff McGill Student Paper Award, 2022

Board Member, 2014–2016

INFORMS Revenue Management & Pricing Conference

– Spotlight Session Discussant, 2022

– Program Committee Member, 2017, 2019, 2022

## INFORMS Applied Probability Society

Markov Lecture Discussant, 2023

INFORMS Applied Probability Society Conference, Session Chair, 2013, 2015, 2017

## INFORMS MSOM Society

Healthcare Management SIG Discussant, 2024

Service Management SIG Discussant, 2024

MSOM Conference Teaching Innovation Session Speaker, 2025

## INFORMS Annual Meeting

Cluster Chair

– MSOM Service Management cluster, 2018

– Revenue Management & Pricing cluster, 2015

Session Chair

– Revenue Management & Pricing cluster, 2012–2024

– Applied Probability Society cluster, 2016, 2017, 2020

– Data Mining cluster, 2020–2022

– MSOM cluster, 2016

– MSOM Service Management cluster, 2019

– MSOM iFORM cluster, 2018, 2020, 2021, 2022

– Decision Analysis Society cluster, 2022

## POMS Annual Conference

Track Chair, Revenue Management & Pricing track, 2025

## YA/EM Congress on Operations Research & Industrial Engineering

Plenary Speaker, 2021

## Judge

George Nicholson Student Paper Competition, 2021, 2022

INFORMS Applied Probability Society Best Student Paper Competition, 2021, 2022

Jeff McGill Student Paper Award, 2025

MSOM Student Paper Competition, 2015–2025

POMS College of Supply Chain Management Best Student Paper Competition, 2024, 2025

## Reviewer

*Management Science, Operations Research, M&SOM, Mathematics of Operations Research, Marketing Science, Production and Operations Management, Decision Analysis, Games and Economic Behavior, INFORMS Journal of Computing, Journal of Applied Probability, European Journal of Operational Research, Operations Research Letters, Stochastic Systems, Naval Research Logistics, Annals of Operations Research, Computers and Operations Research, ACM Transactions on Economics and Computation, IIE Transactions*

MSOM Supply Chain SIG Workshop, 2012, 2015, 2016, 2017

MSOM Service SIG Workshop, 2016–2019

Duke University, Fuqua School of Business

Ph.D. Program Coordinator, Operations Management, 2023–present

Judicial Board Member, 2025–present

Student Learning Group, Operations Management Area Representative, 2025–present

Curriculum Committee Member, 2023–2024

Ph.D. Program Committee Member, Operations Management, 2016–2023

Seminar Organizer, Operations Management, 2016–2018

Website Representative, Operations Management, 2015–2019

University of Chicago, Booth School of Business

Workshop Organizer, Operations Management, 2012–2013

### **Student Supervision and Collaboration**

Duke University, Fuqua School of Business

Yuan-Mao Kao (initial placement: City University of New York, Baruch College, Zicklin School)

Chenghuai Li (initial placement: University of Alberta, Alberta School of Business)

Yuxing Li (initial placement: Johns Hopkins University, Carey Business School)

Vinh Nguyen (initial placement: Marriott International Inc.)

University of Chicago, Booth School of Business

Hongfan Chen (initial placement: Chinese University of Hong Kong Business School)

Yifan Feng (initial placement: National University of Singapore Business School)

Adam Schultz (initial placement: Wealthfront Inc.)

### **Other Professional Experience**

Nomis Solutions, R&D Intern, 2008

Boğaziçi University Flexible Manufacturing Laboratory, Research Assistant, 2003–2006

Unilever, Production Planning Intern, 2005

Kraft Foods, Supply Chain Intern, 2004

Türk Telekom, IT Intern, 2004

### **Other Honors and Awards**

University of Chicago, Booth School of Business, Neubauer Family Faculty Fellowship, 2013–2014

Stanford University, Graduate School of Business, Donald E. Petersen Fellowship, 2008–2010

Boğaziçi University, Graduated summa cum laude, 2007

Ranked first among approximately 600 engineering and math graduates, 2007

Ranked first among all double-major graduates, 2007

President of the Student Representative Council, 2005–2006

Alper Atalay Fellowship (based on academic excellence), 2003–2007

Istanbul Atatürk High School of Science, Graduated summa cum laude, 2001

National Physics Olympiad Marmara Region First Rank, 1999

National Mathematics Olympiad Silver Medal, 1998